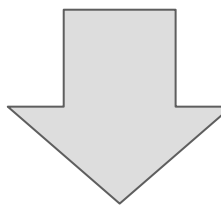


Student Experience in Self-Learning Mechanics Through OER Development

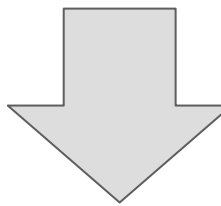
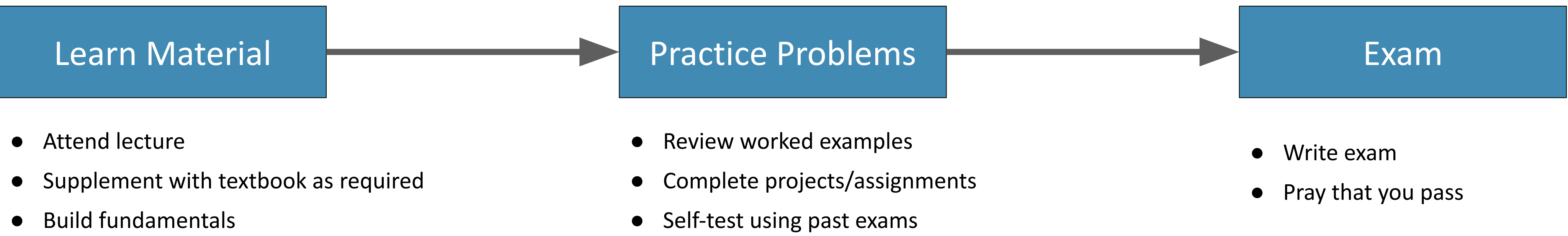
Julian Lapenna and Tyler Wilson

Introduction

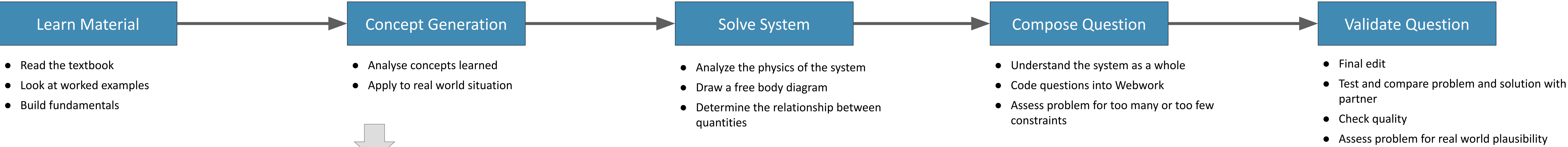
As part of a multi-year project, our co-op role was to develop open educational resources (OER), more specifically practice problems coded for the WeBWork system. Our subject was Rigid Body Dynamics, physics material that neither of us had learned before. This meant that we were simultaneously learning the course material on our own and developing questions for the course which gave us a very different perspective of learning in contrast to the traditional classroom setting.



The Traditional Learning Method



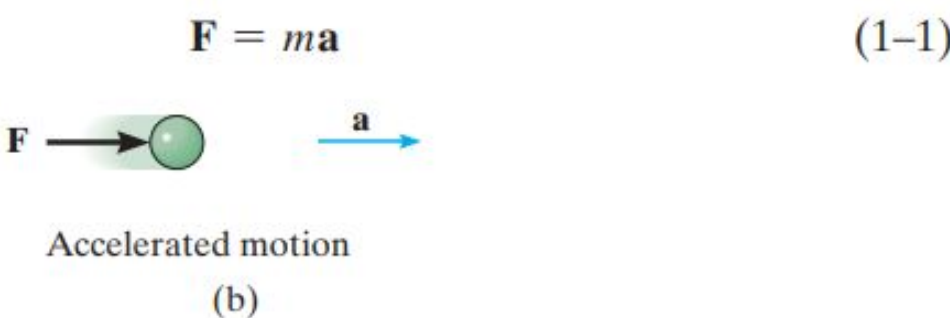
Our Problem Creation Process



Example Process

Example textbook material

Second Law. A particle acted upon by an *unbalanced force* $\mathbf{F}$ experiences an acceleration $\mathbf{a}$ that has the same direction as the force and a magnitude that is directly proportional to the force, Fig. 1-1b.* If $\mathbf{F}$ is applied to a particle of mass m , this law may be expressed mathematically as



Example problem. Red boxes show values that are randomized for students.

This is what a WeBWork question looks like. Here is a sample scenario:

A golf ball of mass $m = 0.04593 \text{ kg}$ is at rest on a golf tee and can be modelled as a particle. A golfer approaches and swings their driver, hitting the golf ball with a force of $F = 11000 \text{ N}$. What is the acceleration of the golf ball just after it has been hit?

$a =$ m/s^2

Submission results: The red box shows the student's submitted answer. In the first example the student is correct. In the second example the student is incorrect and can try again.

Entered	Answer Preview	Result
239495	$\frac{11000}{0.04593}$	correct

The answer above is correct.

Entered	Answer Preview	Result
-239495	$\frac{-11000}{0.04593}$	incorrect

The answer above is NOT correct.

Our Experience

We found that the traditional classroom setting encourages a more algorithmic approach to solving problems, whereas in developing these problems, we were able to get a better understanding of how the system as a whole will behave. This resulted in a much better understanding of the fundamentals and made it much easier to relate what we learned to real-world concepts. We believe that it would be beneficial to students for problem development to be integrated into our traditional classroom settings.

Differences Between Processes

The main differences we identified between the two learning processes are:

1. We looked at problems from **both the instructor and student perspective**. This allowed us to more easily identify particular concepts that we found challenging and address those concepts in the problems that we created.
2. **Working with a partner was essential**. Checking each other's work allowed us to compare solutions and approaches to problems.
3. Each problem must be **solved entirely algebraically** so that a solution can be generated for any combination of values. This led to a better understanding of the system's dependencies as a whole.
4. We **assessed the material wholistically**. When you are working through a homework set you typically do so with the assumption that there is a single correct answer and that you should be able to find it. Comparing this to developing your own problems, you typically start with some physical system that must move according to the laws of Newtonian Mechanics. You must consider how the different components of the system are related and what variables you require in order to create a solvable problem such that the system is not overconstrained or underconstrained.
5. **Creating problems demands more involvement** than answering them. For challenging concepts we devised problems that focused on specific techniques or problems with multiple steps and hints to give them feedback throughout their work.

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